Green University of Bangladesh
Department of Computer Science and Engineering (CSE)

Faculty of Sciences and Engineering

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Lab Report NO #01

Course Title: Database Lab

Course Code: CSE 210 Section: 242-D3

Lab Experiment Name: Design and Implementation of Library Management Database Using SQL Constraints and ALTER Operations.

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Lab Report Status

Marks: …… **Signature:.....**

Comments:..... **Date:.....**

1. TITLE OF THE LAB REPORT EXPERIMENT

Design and Implementation of Library Management Database Using SQL Constraints and ALTER Operations.

2. OBJECTIVES / AIM

- To create a relational database using MySQL.
- To implement Primary Key, Foreign Key, Unique, and Check constraints.
- To use ENUM and DEFAULT values in table design.
- To verify constraint violations (error testing).
- To perform ALTER operations such as add, modify, drop, rename.
- To understand relational integrity in database systems.

3. PROCEDURE / DESIGN/ ALGORITHM

A relational database organizes data into tables (relations) consisting of rows and columns.
Each table must have:

- A Primary Key to uniquely identify records.
- Foreign Keys to maintain relationships between tables.

In this experiment, three tables are created:

- MEMBER
- BOOK
- LOAN (later renamed to BORROW)

Relationships:

- One MEMBER can borrow many BOOKS.
- LOAN connects MEMBER and BOOK using Foreign Keys.

<u>ALGORITHM</u>

Part A: Database Creation
Step 1: Create database library_lab
Step 2: Use the database

Part B: Table Creation
Step 1: Create MEMBER table with constraints
Step 2: Create BOOK table with constraints
Step 3: Create LOAN table with foreign keys

Part C: Data Insertion
Step 1: Insert 5 records into MEMBER
Step 2: Insert 5 records into BOOK
Step 3: Insert 6 records into LOAN

Part D: Constraint Verification
Step 1: Insert duplicate email
Step 2: Insert LOAN with invalid MemberID
Step 3: Insert BOOK with negative price
Step 4: Observe system errors

Part E: ALTER Operations
Step 1: Add Phone column
Step 2: Modify Price datatype
Step 3: Rename Status to Availability
Step 4: Drop Gender column
Step 5: Add named CHECK constraint
Step 6: Rename LOAN to BORROW

4. IMPLEMENTATION

MySQL Query:

```
CREATE DATABASE library_lab;
USE library_lab;

CREATE TABLE MEMBER (
   MemberID INT,
   FullName VARCHAR(100) NOT NULL,
   Email VARCHAR(100) NOT NULL,
   Age INT,
   Gender ENUM('MALE','FEMALE','OTHER'),
   CONSTRAINT PK_MEMBER PRIMARY KEY (MemberID),
   CONSTRAINT UQ_MEMBER_EMAIL UNIQUE (Email),
   CONSTRAINT CHK_MEMBER_AGE CHECK (Age >= 12)
);
```

```
CREATE TABLE BOOK (
   BookID INT,
   Title VARCHAR(150) NOT NULL,
   ISBN VARCHAR(20) NOT NULL,
   Price DECIMAL(6,2),
   Status ENUM('AVAILABLE','BORROWED') DEFAULT 'AVAILABLE',
   CONSTRAINT PK_BOOK PRIMARY KEY (BookID),
   CONSTRAINT UQ_BOOK_ISBN UNIQUE (ISBN),
   CONSTRAINT CHK_BOOK_PRICE CHECK (Price >= 0)
);

CREATE TABLE LOAN (
   LoanID INT,
   MemberID INT,
   BookID INT,
   CONSTRAINT PK_LOAN PRIMARY KEY (LoanID),
   CONSTRAINT FK_LOAN_MEMBER FOREIGN KEY (MemberID) REFERENCES
MEMBER(MemberID),
   CONSTRAINT FK_LOAN_BOOK FOREIGN KEY (BookID) REFERENCES BOOK(BookID)
);

INSERT INTO MEMBER VALUES
(1, 'Rahim Uddin', 'rahim@gmail.com', 20, 'MALE'),
(2, 'Karim Ali', 'karim@gmail.com', 25, 'MALE'),
(3, 'Nusrat Jahan', 'nusrat@gmail.com', 22, 'FEMALE'),
(4, 'Sadia Islam', 'sadia@gmail.com', 18, 'FEMALE'),
(5, 'Tanvir Hasan', 'tanvir@gmail.com', 30, 'MALE');

INSERT INTO BOOK VALUES
(101, 'Database Systems', 'ISBN001', 450.00, 'AVAILABLE'),
(102, 'Java Programming', 'ISBN002', 550.00, 'AVAILABLE'),
(103, 'C Programming', 'ISBN003', 400.00, 'AVAILABLE'),
(104, 'Data Structures', 'ISBN004', 600.00, 'AVAILABLE'),
(105, 'Networking Basics', 'ISBN005', 500.00, 'AVAILABLE');

INSERT INTO LOAN VALUES
(1, 1, 101),
(2, 2, 102),
(3, 3, 103),
(4, 4, 104),
(5, 5, 105),
(6, 1, 102);

INSERT INTO MEMBER VALUES
(6, 'Duplicate User', 'rahim@gmail.com', 28, 'MALE');

INSERT INTO LOAN VALUES
(7, 99, 101);

INSERT INTO BOOK VALUES
```

```
(106, 'Invalid Book', 'ISBN006', -200.00, 'AVAILABLE');

ALTER TABLE MEMBER
ADD Phone VARCHAR(15) NOT NULL DEFAULT 'N/A';

ALTER TABLE BOOK
MODIFY Price DECIMAL(8,2);

ALTER TABLE BOOK
CHANGE Status Availability ENUM('AVAILABLE','BORROWED') DEFAULT 'AVAILABLE';

ALTER TABLE MEMBER
DROP COLUMN Gender;

ALTER TABLE MEMBER
ADD CONSTRAINT CHK_MEMBER_PHONE CHECK (Phone <> '');

RENAME TABLE LOAN TO BORROW;
```

5. TEST RESULT / OUTPUT

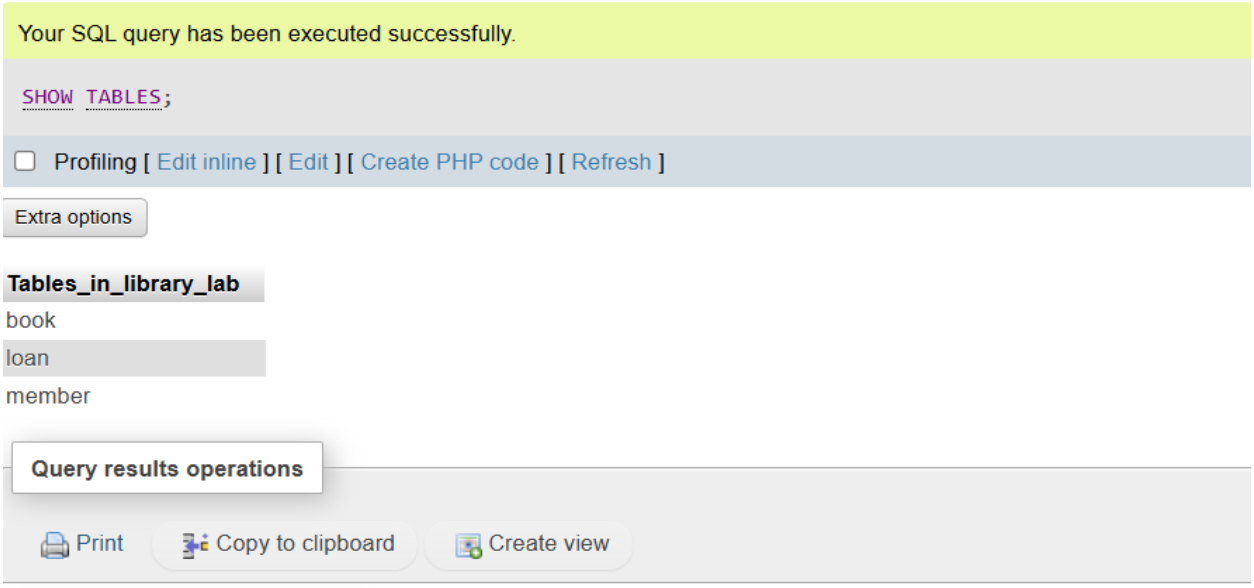

Figure 01 : Create Database and table

Your SQL query has been executed successfully.

```
DESCRIBE BOOK;
```

[Edit inline] [Edit] [Create PHP code]

Extra options

Field	Type	Null	Key	Default	Extra
BookID	int(11)	NO	PRI	NULL	
Title	varchar(150)	NO		NULL	
ISBN	varchar(20)	NO	UNI	NULL	
Price	decimal(6,2)	YES		NULL	
Status	enum('AVAILABLE','BORROWED')	YES		AVAILABLE	

Query results operations

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Figure 02 : Describe table name is Book

Your SQL query has been executed successfully.

```
DESCRIBE MEMBER;
```

[Edit inline] [Edit] [Create PHP code]

Extra options

Field	Type	Null	Key	Default	Extra
MemberID	int(11)	NO	PRI	NULL	
FullName	varchar(100)	NO		NULL	
Email	varchar(100)	NO	UNI	NULL	
Age	int(11)	YES		NULL	
Gender	enum('MALE','FEMALE','OTHER')	YES		NULL	

Query results operations

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Figure 03 : Describe table name is Member

Your SQL query has been executed successfully.

```
DESCRIBE LOAN;
```

[Edit inline] [Edit] [Create PHP code]

Extra options

Field	Type	Null	Key	Default	Extra
LoanID	int(11)	NO	PRI	NULL	
MemberID	int(11)	YES	MUL	NULL	
BookID	int(11)	YES	MUL	NULL	

Query results operations

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Bookmark this SQL query

Figure 04 : Describe table name is Loan

Showing rows 0 - 4 (5 total, Query took 0.0004 seconds.)

```
SELECT * FROM MEMBER;
```

Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]

Show all | Number of rows: 25 Filter rows: Search this table Sort by key: None

Extra options

	MemberID	FullName	Email	Age	Gender
Edit Copy Delete	1	Rahim Uddin	rahim@gmail.com	20	MALE
Edit Copy Delete	2	Karim Ali	karim@gmail.com	25	MALE
Edit Copy Delete	3	Nusrat Jahan	nusrat@gmail.com	22	FEMALE
Edit Copy Delete	4	Sadia Islam	sadia@gmail.com	18	FEMALE
Edit Copy Delete	5	Tanvir Hasan	tanvir@gmail.com	30	MALE

Check all *With selected:* Edit Copy Delete Export

Show all | Number of rows: 25 Filter rows: Search this table Sort by key: None

Figure 05 : Insert values in Member

Showing rows 0 - 4 (5 total, Query took 0.0007 seconds.)

```
SELECT * FROM BOOK;
```

Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]

Show all | Number of rows: 25 Filter rows: Search this table Sort by key: None

Extra options

	BookID	Title	ISBN	Price	Status
Edit Copy Delete	101	Database Systems	ISBN001	450.00	AVAILABLE
Edit Copy Delete	102	Java Programming	ISBN002	550.00	AVAILABLE
Edit Copy Delete	103	C Programming	ISBN003	400.00	AVAILABLE
Edit Copy Delete	104	Data Structures	ISBN004	600.00	AVAILABLE
Edit Copy Delete	105	Networking Basics	ISBN005	500.00	AVAILABLE

Check all *With selected:* Edit Copy Delete Export

Figure 06 : Insert values in Book

Showing rows 0 - 5 (6 total, Query took 0.0004 seconds.)

```
SELECT * FROM LOAN;
```

Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]

Show all | Number of rows: 25 Filter rows: Search this table Sort by key: None

Extra options

	LoanID	MemberID	BookID
Edit Copy Delete	1	1	101
Edit Copy Delete	2	2	102
Edit Copy Delete	3	3	103
Edit Copy Delete	4	4	104
Edit Copy Delete	5	5	105
Edit Copy Delete	6	1	102

Check all *With selected:* Edit Copy Delete Export

Show all | Number of rows: 25 Filter rows: Search this table Sort by key: None

Figure 07 : Insert values in Loan

Error

SQL query: Copy

```
INSERT INTO MEMBER VALUES
(6, 'Duplicate User', 'rahim@gmail.com', 28, 'MALE');
```

MySQL said:

```
#1062 - Duplicate entry 'rahim@gmail.com' for key 'UQ_MEMBER_EMAIL'
```

Figure 08 : Duplicate value Insertion show Error

Your SQL query has been executed successfully.

```
DESCRIBE MEMBER;
```

[Edit inline] [Edit] [Create PHP code]

Extra options

Field	Type	Null	Key	Default	Extra
MemberID	int(11)	NO	PRI	*NULL*	
FullName	varchar(100)	NO		*NULL*	
Email	varchar(100)	NO	UNI	*NULL*	
Age	int(11)	YES		*NULL*	
Phone	varchar(15)	NO		N/A	

Query results operations

Print Copy to clipboard Create view

Figure 09 : Insert column Phone and drop gender from Member

Your SQL query has been executed successfully.

```
DESCRIBE BOOK;
```

[Edit inline] [Edit] [Create PHP code]

Extra options

Field	Type	Null	Key	Default	Extra
BookID	int(11)	NO	PRI	*NULL*	
Title	varchar(150)	NO		*NULL*	
ISBN	varchar(20)	NO	UNI	*NULL*	
Price	decimal(8,2)	YES		*NULL*	
Availability	enum('AVAILABLE','BORROWED')	YES		AVAILABLE	

Query results operations

Print Copy to clipboard Create view

Figure 10 : Modify Price Datatype and Rename Status to Availability from Book

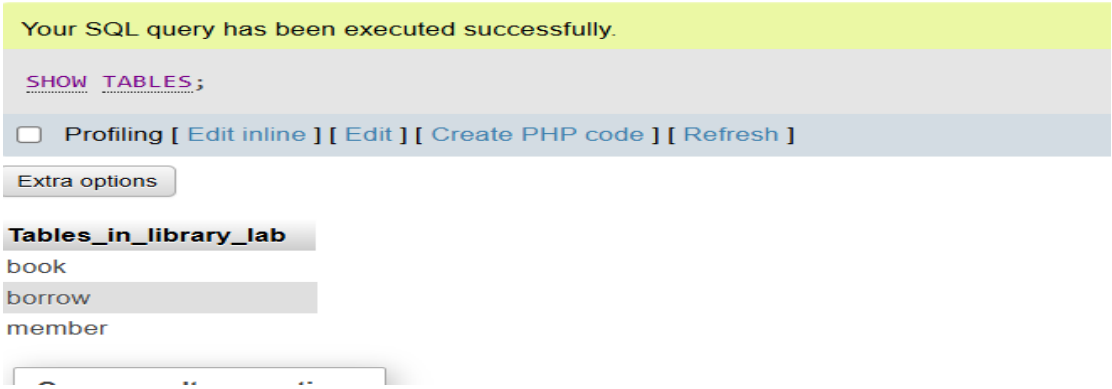

Figure 11 : Rename table name Loan to Borrow

6. ANALYSIS AND DISCUSSION

- Primary Key ensures unique identification of records.
- Foreign Key maintains relationship between MEMBER and BOOK.
- Unique constraint prevents duplicate email and ISBN.
- Check constraint ensures valid Age and Price.
- ENUM restricts allowed values.
- ALTER operations modify database structure dynamically.

7. SUMMARY

In this experiment, a relational database named library_lab was successfully designed and implemented using MySQL. Three interconnected tables MEMBER, BOOK, and LOAN (later renamed to BORROW) were created with properly defined Primary Key, Foreign Key, Unique, and Check constraints to ensure data integrity and consistency. The use of ENUM and DEFAULT values helped control valid data entry. Various constraint violations, such as duplicate email, invalid foreign key, and negative price, were tested to verify that the system correctly enforces rules. Additionally, several ALTER operations were performed, including adding a new column, modifying a datatype, renaming a column, dropping a column, adding a named check constraint, and renaming a table. Overall, this experiment strengthened the understanding of relational database design, constraint implementation, referential integrity, and structural modification of tables in SQL.